

Problem Set 2

Matthew Farah

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Problem 1: *The centralizer of an element.*

Let G be a group and let $a \in G$. The centralizer of a is the set of all elements that commute with a :

$$C(a) := \{g \in G : ga = ag\}$$

(a) In the group S_5 , let $\sigma = (135)$. List all the elements of the centralizer $C(\sigma)$.

Solution:

To find all elements in the centralizer of σ , we must find all elements, $\tau \in S_5$ such that $\sigma\tau = \tau\sigma$.

Helping us in that endeavour is Proposition 2.6., which states that if τ satisfies $\text{supp}(\sigma) \cap \text{supp}(\tau) = \emptyset$, then τ and σ commute. Clearly, σ permutes 1, 3, and 5 while leaving 2 and 4 fixed. So, by definition of the support, $\text{supp}(\sigma) = \{1, 3, 5\}$. Therefore, any τ satisfying $\text{supp}(\tau) \subseteq \{2, 4\}$ must commute with σ . Evidently, there are only two such τ and so we can conclude that at least:

$$\tau = (2, 4), \tau = ()$$

must permute with σ and so belong to the centralizer $C(\sigma)$.

(b) Prove that $C(a) \leq G$ for all $a \in G$.

Solution:

First, let $a \in G$ be arbitrary. To prove that $C(a) \leq G$, we must show that $C(a)$ is a subgroup of G , which we can do by the one-line subgroup test!

The one line subgroup test states that if $xy^{-1} \in C(a)$ for all $x, y \in C(a)$, then $C(a)$ must be a subgroup of G . So, to that end, letting $x, y \in G$ be arbitrary, I will now proceed to show that $(xy^{-1})a = a(xy^{-1})$, hence meaning that $xy^{-1} \in C(a)$ and concluding the proof.

By definition of $C(a)$, we know that $x, y \in a$ implies that $xa = ax$ and $ya = ay$. We also know, since G is a group and $C(a)$ is clearly a subset of G , that y must have an inverse in G . Letting $y^{-1} \in G$ be y 's inverse, we can see that:

$$\begin{aligned} y^{-1}a &= y^{-1}(a(1)) && \text{(By definition of the identity)} \\ &= y^{-1}(a(yy^{-1})) && \text{(By definition of inverses)} \\ &= y^{-1}((ay)y^{-1}) && \text{(By associativity)} \\ &= (y^{-1}(ay))y^{-1} && \text{(By associativity)} \\ &= (y^{-1}(ya))y^{-1} && \text{(Since } y \in C(a)) \\ &= ((y^{-1}y)a)y^{-1} && \text{(By associativity)} \\ &= ((1)a)y^{-1}y^{-1} && \text{(By definition of inverses)} \\ &= ay^{-1} && \text{(By definition of the identity)} \end{aligned}$$

Therefore, $y^{-1}a = ay^{-1}$, implying $y^{-1} \in C(a)$ and thereby making $xy^{-1} \in C(a)$ a corollary because:

$$\begin{aligned}
 (xy^{-1})a &= x(y^{-1}a) && \text{(By associativity)} \\
 &= x(ay^{-1}) && \text{(Since } y \in C(a)) \\
 &= (xa)y^{-1} && \text{(By associativity)} \\
 &= (ax)y^{-1} && \text{(Since } x \in C(a)) \\
 &= a(xy^{-1}) && \text{(By associativity)}
 \end{aligned}$$

Immediately follows. Thus, by the one-line subgroup test, $C(a)$ is indeed a subgroup of G .

(c) Two friends, Goofus and Gallant, are arguing about the following claim:

$$\text{“If } a, b \in G, \text{ then } C(a) \cap C(b) = C(ab)\text{”}$$

Goofus thinks “ $\subseteq$ ” is true, but Gallant thinks “ $\supseteq$ ” is true. Who is correct? Prove one inclusion, and give an explicit counterexample showing that the other inclusion is false in general.

Solution:

I contend, and will proceed to show, that $\subseteq$ is indeed the only correct inclusion. To do this, I intend to formally prove that $(C(a) \cap C(b)) \subseteq C(ab)$ for any arbitrary $a \in G$, for any arbitrary group G , and provide a specific group G with elements $a, b \in G$ such that $(C(a) \cap C(b)) \supseteq C(ab)$ doesn't hold, making Gallant's inclusion false in general.

(I) Prove that $(C(a) \cap C(b)) \subseteq C(ab)$.

To prove this inclusion, I will show that $x \in (C(a) \cap C(b))$ implies that $x \in C(ab)$. So, let any such x be arbitrary.

By definition of set intersection, we know that $x \in (C(a) \cap C(b))$ implies $x \in C(a)$ and $x \in C(b)$, which by definition of the centralizer, means that $xa = ax$ and $xb = bx$. To prove that $x \in C(ab)$, we need only show that $x(ab) = (ab)x$, however from our preceding observations, we can infer that:

$$\begin{aligned}
 x(ab) &= (xa)b && \text{(By associativity)} \\
 &= (ax)b && \text{(Since } x \in C(a)) \\
 &= a(xb) && \text{(By associativity)} \\
 &= a(bx) && \text{(Since } x \in C(b)) \\
 &= (ab)x && \text{(By associativity)}
 \end{aligned}$$

And hence, x belongs to the centralizer $C(ab)$. Therefore, since $x \in (C(a) \cap C(b))$ implies $x \in C(ab)$ for arbitrary x , we have that $(C(a) \cap C(b)) \subseteq C(ab)$, proving the inclusion.

(II) Show $(C(a) \cap C(b)) \supseteq C(ab)$ is false in general.

To show that this inclusion is generally false, we need only observe a single group G with specific elements $a, b \in G$ such that $x \in C(ab)$ but $x \notin (C(a) \cap C(b))$.

Problem 2: *True or false?*

Determine which of the following statements are true. Supply a proof or counterexample as necessary. Note that your examples should be fully justified: you must prove that your example has the necessary properties.

- (a) “ D_5 has exactly one subgroup of order 5.”

Solution:

- (b) “Every proper subgroup of A_4 is cyclic.”

Solution:

- (c) “ $\mathbb{Q}$ is a cyclic group under addition.”

Solution:

- (d) “Let G be a group. Then $Z(G)$ is cyclic.”

Solution:

Problem 3: *The relationship between $\text{ord}(x)$ and $\text{ord}(x^m)$.*

Let G be a group, and let x be an element of G of finite order $n = \text{ord}(x)$. Let m be an integer which is coprime to n . Prove that $\text{ord}(x^m) = n$.

Hint: You may use the following lemma without proof:

Suppose that a, b, c are three positive integers such that a, b are coprime and a divides bc . Then, a divides c .

Solution:

Problem 4: *The relationship between $\text{ord}(x)$, $\text{ord}(y)$, and $\text{ord}(xy)$.*

This is a follow-up to a question from the May 27th tutorial.

Let G be a group, and let $x, y \in G$ be elements of finite orders m and n respectively. Suppose that $xy = yx$ and $\langle x \rangle \cap \langle y \rangle = 1$. Prove that:

$$\text{ord}(xy) = \text{lcm}(m, n)$$

Solution:

Problem 5: *Is a direct product of cyclic groups still cyclic?*

(a) Let A and B be two groups. Prove that if $A \times B$ is cyclic, then so are A and B .

Solution:

(b) Suppose that A, B are finite, cyclic groups of orders m and n respectively, and assume that $\gcd(m, n) = 1$. Prove that $A \times B$ is cyclic.

Solution:

Problem 6: *Let's get working with dihedral groups!*

In this problem, you will be investigating the dihedral group D_{17} : the group of adjacency-preserving permutations of a heptadecagon. You will see that you can work with D_{17} even if you don't know anything about permutations or heptadecagons — all you need is algebra.

There are two key elements of this group: the clockwise rotation r , and a reflection s . They obey the following four algebraic rules:

- *Rule #1:* There are thirty-four elements in D_{17} , split into two types:
 - Seventeen **rotations**: $1, r, r^2, \dots, r^{16}$; and
 - Seventeen **reflections**: $s, sr, sr^2, \dots, sr^{16}$.
- *Rule #2:* $\text{ord}(r) = 17$.
- *Rule #3:* $\text{ord}(s) = 2$.
- *Rule #4:* $rs = sr^{-1}$.

Answer each question using *only* these four rules. You do not need to prove these rules.

(a) Prove that $r^k s = sr^{-k}$.

Hint: Induction, with Rule #4 being the base case.

Solution:

(b) Show that the elements $r, r^2, r^3, \dots, r^{16}$ all have order 17, and that the elements $s, sr, sr^2, \dots, sr^{16}$ all have order 2.

Solution:

(c) Find the conjugacy classes of r and s .

Hint: To find the conjugacy class of r , technically you'd have to compute grg^{-1} for each of the thirty-four elements g in D_{17} . That's a lot of work! Is there a simpler way to handle several cases at once?

Solution:

(d) Show that $\langle r \rangle$ is a normal subgroup of D_{17} .

Hint: There is a useful exercise in the June 3rd tutorial.

Solution:

Problem 7: *The diagonal subgroup*

Let G be a group. The **diagonal** of G is the following subset of $G \times G$.

$$\Delta := \{ (x, y) \in G \times G : x = y \}$$

(a) Prove that $\Delta \leq G \times G$.

Solution:

(b) Prove that ...

Solution:

Problem 8: *Normal subgroups are friends with everyone*

Let G be a group, and let H and K be two subgroups of G . In this problem, we introduce the following definition: H is said to be friends with K if the product set HK is a subgroup of G . (The definition of “product set” is in the May 20th tutorial).

- (a) Give an example of a group G containing two subgroups H and K such that H is not friends with K .

Solution:

- (b) Prove that if H is a subgroup of G , then $HH = H$. Conclude that H is always friends with itself.

Solution:

- (c) Prove that if H is friends with K , then $HK = KH$. Conclude that K must be friends with H too.

Solution:

- (d) Suppose that H, K , and L are three subgroups of G . Suppose that H is friends with K , and K is friends with L . Does it follow that H is friends with L too? Either prove it or provide an explicit counter-example.

Solution:

- (e) Let N be a subgroup of G . Prove that if N is a normal subgroup of G , then N is friends with every subgroup of G .

Solution: